AZ-104 Lab 02b – Manage Governance via Azure Policy

Lab Objective

This lab explains how Azure Policy, resource tags, and resource locks are used to enforce governance and compliance in Azure. It focuses on enforcing standards, remediating non-compliant resources, and protecting critical resources from accidental changes.

Core Governance Concepts

Governance in Azure ensures that resources follow organizational standards. Azure Policy enforces rules, tags provide metadata for visibility, and resource locks protect resources after deployment.

Azure Tags

Tags are key-value metadata pairs applied to Azure resources or resource groups. They are commonly used for cost tracking, ownership, project identification, and automation.

Azure Policy

Azure Policy evaluates resources against defined rules to ensure compliance. Policies can deny deployments, audit resources, or automatically remediate issues.

Azure Policy Effects

- Deny – Blocks creation of non-compliant resources.

- Audit – Allows creation but marks resource as non-compliant.

- Modify – Automatically fixes non-compliant resources.

- DeployIfNotExists – Deploys required configuration if missing.

Resource Locks

Resource locks prevent accidental deletion or modification of resources. Locks override RBAC permissions, even for owners.

Lab Tasks – Detailed Explanation

Task 1: Assign Tags

In this task, a Cost Center tag is applied to a resource group. This ensures all resources can later inherit or be validated against this tag.

Task 2: Enforce Tagging (Deny)

The 'Require a tag and its value' policy enforces tagging by blocking resource creation if the required tag is missing. This is a pre-deployment governance control.

Task 3: Apply Tagging (Modify)

The 'Inherit a tag from the resource group if missing' policy automatically adds missing tags to existing and new resources using a remediation task.

Task 4: Resource Locks

A delete lock is applied to prevent accidental deletion of the resource group, demonstrating post-deployment protection.

Interview-Ready Questions and Answers

Q: What is Azure Policy?

A: Azure Policy enforces organizational standards by evaluating resources against compliance rules.

Q: Difference between Azure Policy and Azure RBAC?

A: Azure Policy enforces what is allowed, while Azure RBAC controls who can access resources.

Q: What are remediation tasks?

A: Remediation tasks fix existing non-compliant resources for Modify or DeployIfNotExists policies.

Q: Do resource locks override RBAC?

A: Yes, resource locks override RBAC permissions, even for owners.

Q: Why use tags in Azure?

A: Tags provide metadata for cost management, ownership, and reporting.

Practical Commands

Resource Locks – PowerShell

```
New-AzResourceLock -LockName rg-lock -LockLevel CanNotDelete -ResourceGroupName az104-rg2
```

```
Remove-AzResourceLock -LockName rg-lock -ResourceGroupName az104-rg2
```

Resource Locks – Azure CLI

```
az lock create --name rg-lock --lock-type CanNotDelete --resource-group az104-rg2
```

```
az lock delete --name rg-lock --resource-group az104-rg2
```